

# LSD, Psilocybin and Ketamine in the Long COVID / ME-CFS Community

**A within-patient analysis of self-reported treatment outcomes.**

Source: `data/full_corpus_2026-07-31/` — 69,161 patient records extracted from 15 long-COVID / ME-CFS subreddits, 204,417 `treatment_outcome` triples.

This notebook is statistics only. Methods, Results, Conclusion. Narrative prose is deliberately left out.

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## The question, and why the obvious analysis is wrong

The community treats these three substances as one family. A naive read of this corpus says psilocybin is reported as helping 81% of the time against a 67% corpus-wide average — but the people who write about psychedelics are not the corpus average. Section 2.3 measures the gap: they are no more physically disabled, but their records carry twice the rate of depression and two and a half times the rate of suicidal ideation, and they name a median of 7 treatments against everyone else's 3. The comparison measures *who reports*, not *what the drug did*.

So the comparator here is **the patient themselves**: every psychedelic outcome is compared against the same patient's outcomes for their own other treatments, with patient fixed effects. The two documented biases in this dataset — an extractor that over-calls positive, and stacked treatments inheriting a collective outcome — apply to both arms inside the same record and largely cancel.

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## 1. Methods

### 1.1 Data source and unit of analysis

Input is `records_covidlonghaulers_v2.json`, the run's authoritative dataset, rather than the `records.csv` convenience table, which is documented as lossy.

One record is one **patient**, not one post — every post and comment by an author across the whole corpus was concatenated and extracted in a single LLM call (deepseek-v4-flash, temperature 0). Authors with fewer than 3 text segments were dropped.

The analysis unit below is one **outcome triple**: `drug : outcome : symptom`, e.g. `"psilocybin: helped: brain fog"`. Outcome is a closed 5-value vocabulary.

```
In [1]: import sys, warnings
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

sys.path.insert(0, str(Path.cwd()))
import psychedelics as P

# The conditional-logit estimator warns each time it drops concordant strata
# That is the design, not a problem -- see 4.1. Silenced to keep output readable
warnings.filterwarnings("ignore", message="Dropped .* groups")

plt.rcParams.update({
    "figure.dpi": 120, "font.size": 9, "axes.grid": True,
    "grid.alpha": 0.25, "axes.spines.top": False, "axes.spines.right": False
})

records = P.load_records()
df = P.build_outcome_table(records)  # asserts every triple parses
df = P.add_prep(df)

print(f"patients in corpus      {len(records):,}")
print(f"outcome triples parsed    {len(df):,}")
print(f"unparseable triples        0 (asserted by build_outcome_table)")
```

```
patients in corpus      69,161
outcome triples parsed  204,417
unparseable triples     0 (asserted by build_outcome_table)
```

## 1.2 Cohort definition

Nothing in this dataset is canonicalized — `ldn`, `low dose naltrexone` and `low-dose naltrexone` are three separate strings among 18,101. Arms are therefore assigned by regex over the **drug slot only**, never over the whole record.

| arm        | pattern                                                                                                                                                                                                                                              |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| psilocybin | <code>psilocyb</code> , <code>psilocin</code> , <code>magic mushroom</code> , <code>shroom(s)</code> , <code>psychedelic mushroom</code> — <b>minus</b> functional mushrooms (lion's mane, reishi, chaga, cordyceps, turkey tail, maitake, shiitake) |
| ketamine   | <code>ketamine</code> , <code>esketamine</code> , <code>spravato</code> — both of the first two are required, since the word boundary in <code>\bketamine\b</code> does not match <code>esketamine</code>                                            |
| LSD        | <code>lsd</code> , <code>lysergic</code> , <code>1p-lsd</code> , <code>ald-52</code> , <code>acid tabs</code> , <code>acid trip</code> , <code>microdose acid</code>                                                                                 |

Bare **acid** never matches. **alpha lipoic acid**, **ascorbic acid**, **folic acid** and **acid reflux** are all common here and would swamp an arm this small. Verified below.

```
In [2]: must_not_match_lsd = [
        "alpha lipoic acid", "r-alpha lipoic acid", "ascorbic acid", "folic acid",
        "amino acids", "hyaluronic acid", "acid reflux", "stomach acid", "fulvic acid",
    ]
    for s in must_not_match_lsd:
        assert P.classify_drug(s) != "lsd", f"FALSE POSITIVE: {s!r}"

    must_match = {
        "lsd": "lsd", "lp-lsd": "lsd", "acid tabs": "lsd", "microdosing lsd": "lsd",
        "psilocybin": "psilocybin", "magic mushrooms": "psilocybin", "shrooms": "psilocybin",
        "ketamine": "ketamine", "spravato": "ketamine", "esketamine": "ketamine",
        "lion's mane": "other", "reishi mushroom": "other",
    }
    for s, want in must_match.items():
        got = P.classify_drug(s)
        assert got == want, f"{s!r} -> {got}, expected {want}"

    print(f"drug-slot classifier: {len(must_not_match_lsd)} false-positive guards and {len(must_match)} assignments all pass")
```

drug-slot classifier: 9 false-positive guards and 12 assignments all pass

## 1.3 Denominators: naming a drug vs reporting how it went

Two different cohorts exist and they must not be confused. A patient can name psilocybin in **medications** without ever saying what it did. Prior work in this repo reported **538** psilocybin patients using the any-mention definition; only a subset of those carry a scored outcome triple, and the triple cohort is what every analysis below uses.

```
In [3]: mentions = P.mention_cohort(records)          # named anywhere: reconciles w
        cohort = P.cohort_counts(df)                 # has a scored outcome triple

        recon = (mentions.set_index("drug_class")
                  .join(cohort[["patients", "mentions"]])
                  .rename(columns={"patients": "patients_with_outcome",
                                   "mentions": "outcome_triples"}))
                  .loc[list(P.DRUGS)])
        recon["pct_with_outcome"] = 100 * recon.patients_with_outcome / recon.patients
        display(recon)

        print("Prior psilocybin analysis reported 538 any-mention patients; reproduced below")
```

|            | patients_mentioning | patients_with_outcome | outcome_triples | pct_with_outcome |
|------------|---------------------|-----------------------|-----------------|------------------|
| drug_class |                     |                       |                 |                  |
| psilocybin | 538                 | 427                   | 717             | 79.36            |
| ketamine   | 525                 | 422                   | 677             | 80.38            |
| lsd        | 94                  | 67                    | 92              | 71.27            |

Prior psilocybin analysis reported 538 any-mention patients; reproduced above.

## 1.4 Excluded ambiguity buckets

Two string families cannot be assigned safely and are counted, reported, then **excluded from every inferential fit**:

- bare `mushroom` / `mushroom blend` / `mushroom complex` with no functional-mushroom name and no explicit psilocybin word,
- bare `acid` strings that are not a known non-psychedelic acid compound.

Section 5.2 re-runs the primary model with both folded back in, to bound how much the exclusion could matter.

```
In [4]: display(cohort.loc[[c for c in cohort.index if c.startswith("ambiguous")]])
```

|                    | patients | mentions | pct_of_corpus_patients |
|--------------------|----------|----------|------------------------|
| drug_class         |          |          |                        |
| ambiguous_mushroom | 139      | 182      | 0.20098                |
| ambiguous_acid     | 83       | 89       | 0.12001                |

## 1.5 Endpoint

Primary endpoint is **helped vs not-helped** (binary). The 5-level vocabulary is too unbalanced corpus-wide (`mixed` 1.5%, `unknown` 1.0%) to support a multinomial fit at LSD's n.

`no_effect` is reported as a secondary endpoint and is arguably the more diagnostic of the two: a positively-biased extractor inflates `helped` far more readily than it suppresses `no_effect`, and a genuinely inert intervention should produce a *high* no-effect rate.

## 1.6 Pre-registered tests, power, and direction of bias

Seven tests, declared before results, Holm-corrected as one family (section 6).  
Everything else in this notebook is exploratory and is reported **without** p-values.

1–3. psilocybin vs ketamine, psilocybin vs LSD, ketamine vs LSD (head-to-head) 4–6.  
each drug vs the same patients' own other treatments (within-patient) 7. psychedelic x  
symptom-class interaction (mood/cognitive vs energy/PEM)

Two things to fix in mind before reading any estimate:

- **LSD is underpowered by design, not by accident.** Its minimum detectable difference is computed below. A null LSD result is uninformative, not evidence of no effect.
- **Noise here is conservative.** Repeat extraction passes over identical text agree on only 58.8% of filled values. Non-differential misclassification biases a within-patient contrast *toward* the null, so an effect that survives section 4 is an underestimate, not an artifact.

```
In [5]: for d in P.DRUGS:
        n = int(cohort.loc[d, "mentions"])
        print(f"{d:11s} n={n:4d} mentions -> minimum detectable risk difference
              f"{100*P.mde(0.68, n):.1f} pp (80% power, alpha .05, vs a 68% base
```

```
psilocybin n= 717 mentions -> minimum detectable risk difference 6.9 pp (8
0% power, alpha .05, vs a 68% baseline)
ketamine   n= 677 mentions -> minimum detectable risk difference 7.1 pp (8
0% power, alpha .05, vs a 68% baseline)
lsd        n=  92 mentions -> minimum detectable risk difference 19.3 pp (8
0% power, alpha .05, vs a 68% baseline)
```

## 1.7 Symptom classes

Fixed before any outcome was inspected. Only the first two are tested; **pain** and  
**other** are descriptive.

```
In [6]: sym = (df[df.drug_class.isin(P.DRUGS)]
          .groupby(["drug_class", "symptom_class"]).size().unstack(fill_value=0)
          display(sym[["mood_cognitive", "energy_pem", "pain", "other", "unspecified"]])
```

| symptom_class | mood_cognitive | energy_pem | pain | other | unspecified |
|---------------|----------------|------------|------|-------|-------------|
| drug_class    |                |            |      |       |             |
| ketamine      | 240            | 58         | 105  | 169   | 105         |
| lsd           | 22             | 16         | 10   | 29    | 15          |
| psilocybin    | 283            | 72         | 48   | 218   | 96          |

## 1.8 Limits that no amount of statistics repairs

- **Cohort-level claims only.** Two identical extraction passes agreed on the field set for 36.5% of records. Aggregate rates are stable to 0.8pp; a single record is one draw. No case studies, no per-patient narratives.
- **No time axis.** A record collapses years of posting into one document; a 2021 trip and a 2026 trip sit in it unordered. No before/after, no dose-response over time, no trends.
- **No denominator for silent failure.** People post about treatments that did *something*. A drug tried once and quietly abandoned never appears. The within-patient design controls for *reporter* bias; it cannot control for *selection into reporting*.
- **Route asymmetry is structural.** Ketamine is substantially clinic-administered; psilocybin and LSD are self-sourced. Any ketamine-vs-others difference is confounded by supervision, screening and expectancy. See section 4.4.
- **Not the drug-sentiment pipeline.** `treatment_outcome` comes from variable extraction. `src/run_sentiment_pipeline.py` has never been run on this corpus; these rates are not comparable to `sentiment` numbers from other PatientPunk runs.

## 2. Results — descriptive

### 2.1 Raw outcome rates

Every rate carries a Wilson 95% CI and its denominator. The corpus-wide line is a **descriptive reference only** — it is not a control, because the patients in these arms differ systematically from the corpus average on exactly the traits that drive reporting.

```
In [7]: drugs_df = df[df.drug_class.isin(P.DRUGS)]

helped = P.rate_table(drugs_df, col="helped").set_index("drug_class")
noeff = P.rate_table(drugs_df, col="no_effect").set_index("drug_class")

summary = pd.DataFrame({
    "patients": helped.patients, "mentions": helped.n,
    "helped": helped.rate, "helped_lo": helped.ci_lo, "helped_hi": helped.ci_hi,
    "no_effect": noeff.rate, "no_effect_lo": noeff.ci_lo, "no_effect_hi": noeff.ci_hi
})
display(summary.style.format("{:.3f}", subset=[c for c in summary.columns
                                                if c not in ("patients", "mer

print(f"corpus-wide reference (n={P.CORPUS_MENTIONS:,} mentions, NOT a contr
      f"helped {P.CORPUS_BASELINE['helped']:.3f}  "
      f"no_effect {P.CORPUS_BASELINE['no_effect']:.3f}  "
      f"worsened {P.CORPUS_BASELINE['worsened']:.3f}")
```

|            | patients | mentions | helped | helped_lo | helped_hi | no_effect | no_effect_lo |
|------------|----------|----------|--------|-----------|-----------|-----------|--------------|
| drug_class |          |          |        |           |           |           |              |
| psilocybin | 427      | 717      | 0.810  | 0.780     | 0.837     | 0.089     | 0.071        |
| ketamine   | 422      | 677      | 0.694  | 0.659     | 0.728     | 0.157     | 0.131        |
| lsd        | 67       | 92       | 0.696  | 0.595     | 0.780     | 0.098     | 0.052        |

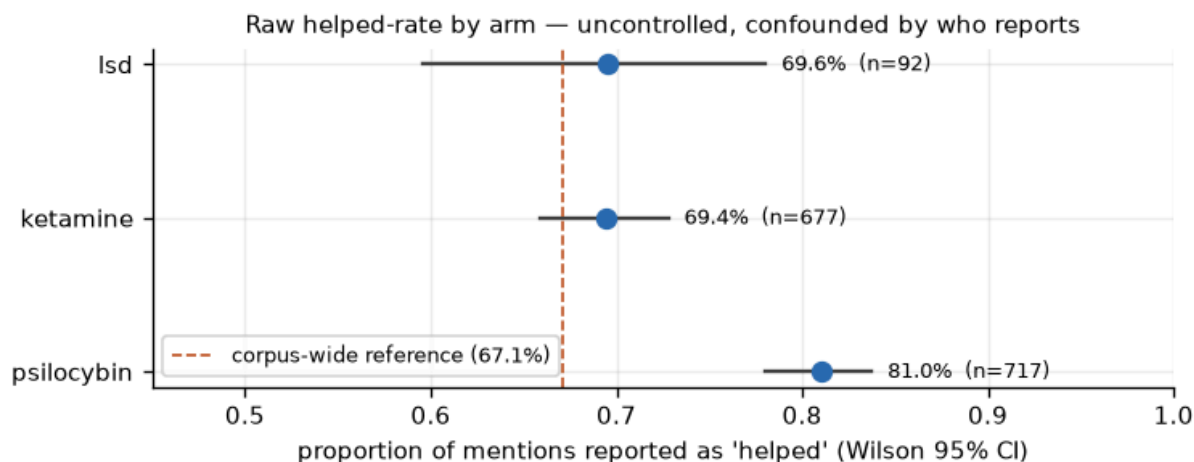
corpus-wide reference (n=204,417 mentions, NOT a control): helped 0.671 no\_effect 0.143 worsened 0.161

```
In [8]: full = (drugs_df.groupby(["drug_class", "outcome"]).size()
              .unstack(fill_value=0).reindex(columns=P.OUTCOMES, fill_value=0))
display(full)
display((100 * full.div(full.sum(axis=1), axis=0)).round(1))
```

|            | outcome | helped | worsened | no_effect | mixed | unknown |
|------------|---------|--------|----------|-----------|-------|---------|
| drug_class |         |        |          |           |       |         |
| ketamine   |         | 470    | 85       | 106       | 10    | 6       |
| lsd        |         | 64     | 18       | 9         | 1     | 0       |
| psilocybin |         | 581    | 62       | 64        | 9     | 1       |

|            | outcome | helped | worsened | no_effect | mixed | unknown |
|------------|---------|--------|----------|-----------|-------|---------|
| drug_class |         |        |          |           |       |         |
| ketamine   |         | 69.4   | 12.6     | 15.7      | 1.5   | 0.9     |
| lsd        |         | 69.6   | 19.6     | 9.8       | 1.1   | 0.0     |
| psilocybin |         | 81.0   | 8.6      | 8.9       | 1.3   | 0.1     |

```
In [9]: fig, ax = plt.subplots(figsize=(6.4, 2.6))
order = list(P.DRUGS)
y = np.arange(len(order))
for i, d in enumerate(order):
    r = helped.loc[d]
    ax.plot([r.ci_lo, r.ci_hi], [i, i], color="#444", lw=1.4, zorder=2)
    ax.plot(r.rate, i, "o", ms=7, color="#2b6cb0", zorder=3)
    ax.annotate(f"{r.rate:.1%} (n={int(r.n)}", (r.ci_hi, i),
               xytext=(6, 0), textcoords="offset points", va="center", font
ax.axvline(P.CORPUS_BASELINE["helped"], color="#c05621", ls="--", lw=1,
           label=f"corpus-wide reference ({P.CORPUS_BASELINE['helped']:.1%})
ax.set_yticks(y, order)
ax.set_xlabel("proportion of mentions reported as 'helped' (Wilson 95% CI)")
ax.set_xlim(0.45, 1.0)
ax.set_title("Raw helped-rate by arm – uncontrolled, confounded by who repor
ax.legend(fontsize=8, loc="lower left")
plt.tight_layout(); plt.show()
```

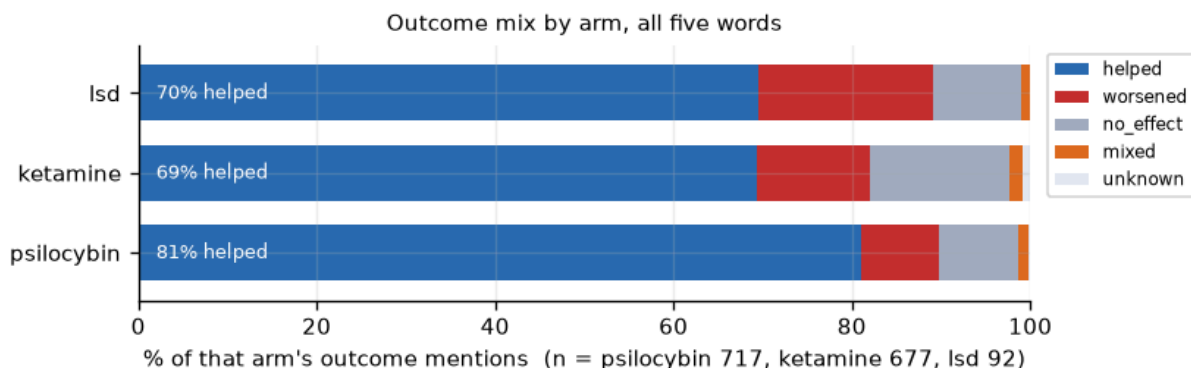


## 2.2 The full outcome vocabulary

The binary endpoint hides three of the five outcome words. Here is the whole vocabulary, as each arm's share of its own mentions.

**worsened** is not a small category and it splits the arms further apart than **helped** does. Psilocybin's share is 8.6%, roughly half the corpus-wide 16.1%. Ketamine's is 12.6%. LSD's is 19.6% — above the corpus, on 18 mentions, which is too few to lean on. Section 4.5 reads those mentions individually rather than as a rate.

```
In [10]: mix = (100 * full.div(full.sum(axis=1), axis=0)).loc[list(P.DRUGS), list(P.C
ax = mix.plot(kind="barh", stacked=True, figsize=(7, 2.3), width=0.7,
              color=["#2b6cb0", "#c53030", "#a0aec0", "#dd6b20", "#e2e8f0"])
for i, d in enumerate(mix.index):
    ax.annotate(f"{mix.loc[d, 'helped']:.0f}% helped", (2, i), va="center",
               fontsize=8, color="white")
ax.set_xlabel(f"% of that arm's outcome mentions "
              f"(n = {'', '.join(f'{d} {int(full.loc[d].sum())}' for d in P.D
ax.set_ylabel("")
ax.set_title("Outcome mix by arm, all five words", fontsize=9)
ax.legend(fontsize=7, bbox_to_anchor=(1.01, 1), loc="upper left")
plt.tight_layout(); plt.show()
```



## 2.3 Who names a psychedelic

Both groups below are patients with at least one scored outcome triple, so they differ in nothing structural except whether a psychedelic appears in their record.

The result is not the one the introduction assumed. On physical severity the two groups are the same: the functional-status tiers sit within a couple of points of each other, and so do the illness trajectories. Psychedelic-naming patients are not more disabled.

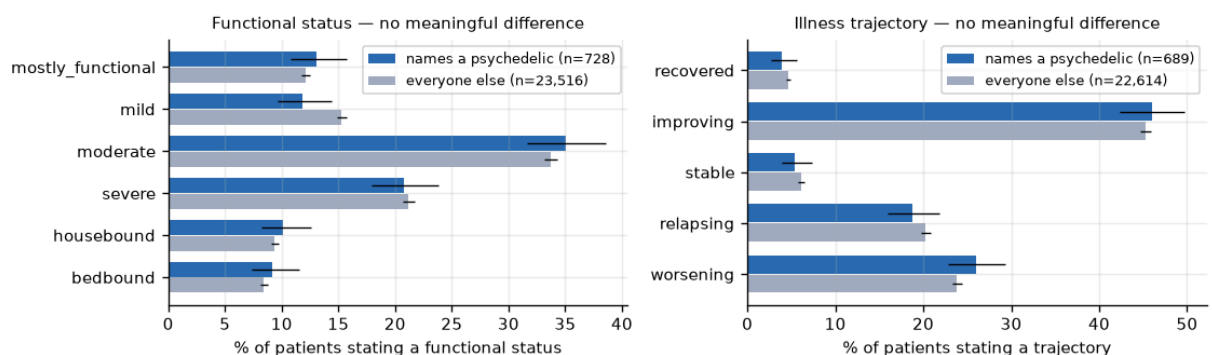
They are, however, carrying a much heavier psychiatric load. Depression appears in their records twice as often, suicidal ideation two and a half times as often, anhedonia four times as often. That single difference explains most of what follows: the indication map in 2.7, the symptom-class interaction in 4.3, and why the raw rates in 2.1 cannot be read as drug effects.

```
In [11]: def wilson_err(pct, n):
    est, lo, hi = P.wilson(int(round(pct / 100 * n)), n)
    return 100 * (est - lo), 100 * (hi - est)

def profileBars(ax, tab, denoms, xlabel):
    y = np.arange(len(tab))[:-1]
    for j, g in enumerate(tab.columns):
        err = np.array([wilson_err(v, denoms[g]) for v in tab[g]]).T
        ax.barh(y + (0.5 - j) * 0.38, tab[g], height=0.36, xerr=err,
                error_kw={"lw": 0.8}, color=["#2b6cb0", "#a0aec0"][j],
                label=f"{g} (n={denoms[g]:,})")
    ax.set_yticks(y, tab.index)
    ax.set_xlabel(xlabel)
    ax.legend(fontsize=7.5)

tier, tier_n = P.field_profile(records, df, "functional_status_tier", P.FUNC
traj, traj_n = P.field_profile(records, df, "illness_trajectory", P.TRAJECTO

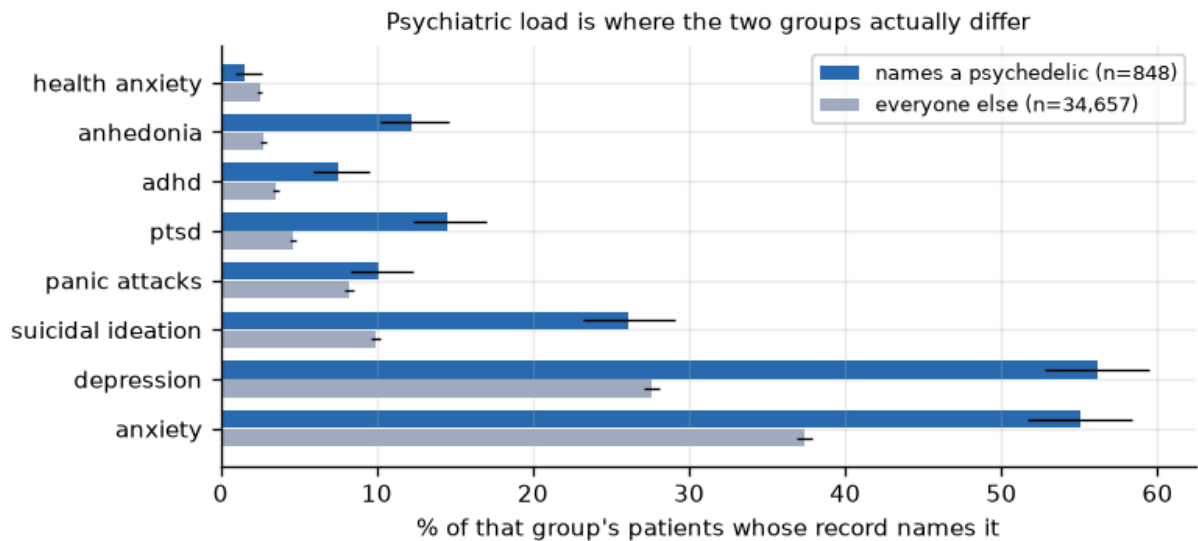
fig, axes = plt.subplots(1, 2, figsize=(9.5, 2.9))
profileBars(axes[0], tier, tier_n, "% of patients stating a functional stat
profileBars(axes[1], traj, traj_n, "% of patients stating a trajectory")
axes[0].set_title("Functional status — no meaningful difference", fontsize=9
axes[1].set_title("Illness trajectory — no meaningful difference", fontsize=
plt.tight_layout(); plt.show()
```



```
In [12]: mh, mh_n = P.field_profile(records, df, "mental_health", conditional=False)
mh = mh.head(8).iloc[:-1]

fig, ax = plt.subplots(figsize=(6.6, 3.1))
profile_bars(ax, mh, mh_n, "% of that group's patients whose record names it")
ax.set_title("Psychiatric load is where the two groups actually differ", fontweight='bold')
plt.tight_layout(); plt.show()

print("Denominators are all patients in each group, not just those discussing mental health. A record silent on depression is a record that did not mention it, which is not the same as a patient who does not have it.")
```



Denominators are all patients in each group, not just those discussing mental health. A record silent on depression is a record that did not mention it, which is not the same as a patient who does not have it.

## 2.4 The stack these substances sit in

Nobody in this cohort tries a psychedelic first. Patients who name one report a median of 7 distinct treatments; everyone else reports 3. A quarter of them name 13 or more.

The company these substances keep is mostly the community's standard repertoire — LDN, cannabis, gabapentin, antihistamines, magnesium. Psychedelics are late entries in a long sequence of attempts, which is worth holding onto: a treatment tried after six failures is being judged against six failures.

```
In [13]: ss = P.stack_sizes(df)
print(ss.groupby("group").n_treatments.describe()[["count", "25%", "50%", "75%", "max"]])

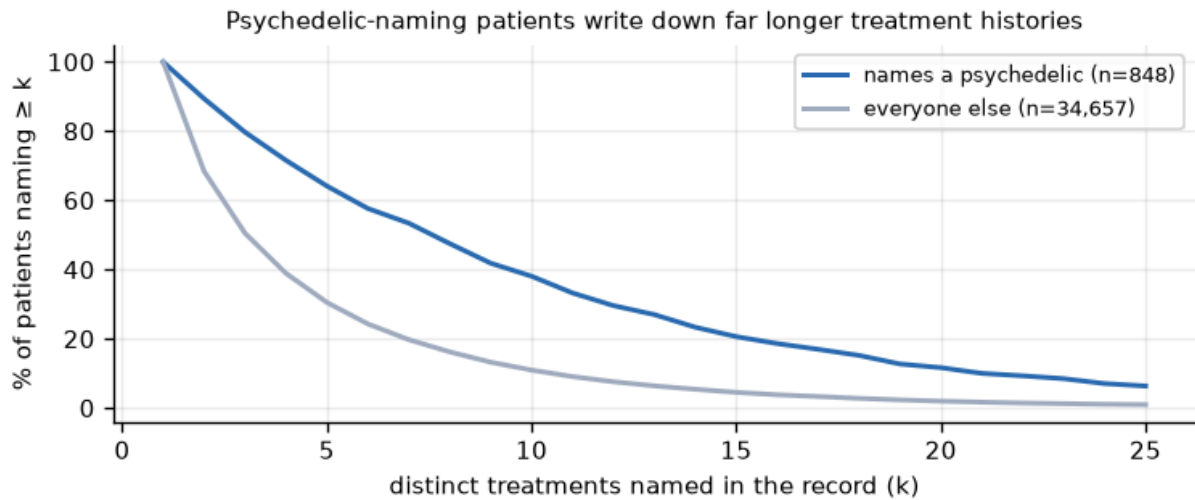
fig, ax = plt.subplots(figsize=(6.4, 2.8))
ks = np.arange(1, 26)
for g, color in [("names a psychedelic", "#2b6cb0"), ("everyone else", "#a0a0a0")]:
    v = ss.loc[ss.group == g, "n_treatments"]
    ax.plot(ks, [100 * (v >= k).mean() for k in ks], color=color, lw=1.8,
```

```

        label=f"{g} (n={len(v):,})")
    ax.set_xlabel("distinct treatments named in the record (k)")
    ax.set_ylabel("% of patients naming ≥ k")
    ax.set_title("Psychedelic-naming patients write down far longer treatment hi
                fontsize=9)
    ax.legend(fontsize=8)
    plt.tight_layout(); plt.show()

```

|                     | count   | 25% | 50% | 75%  | max   |
|---------------------|---------|-----|-----|------|-------|
| group               |         |     |     |      |       |
| everyone else       | 34657.0 | 1.0 | 3.0 | 5.0  | 117.0 |
| names a psychedelic | 848.0   | 3.0 | 7.0 | 13.0 | 140.0 |

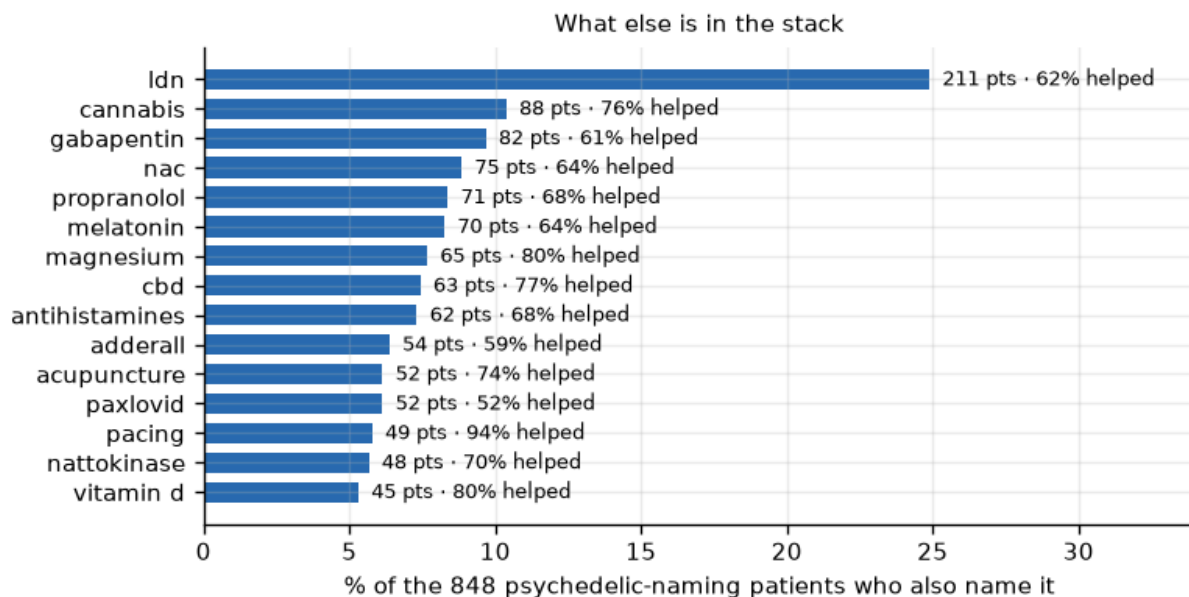


```

In [14]: co_tx = P.co_treatments(df, 15).iloc[::-1]

fig, ax = plt.subplots(figsize=(6.6, 3.4))
ax.barh(np.arange(len(co_tx)), co_tx.pct_of_psychedelic_patients, color="#2b
        height=0.7)
for i, (n, h) in enumerate(zip(co_tx.patients, co_tx.helped)):
    ax.annotate(f"{n} pts · {h:.0%} helped", (co_tx.pct_of_psychedelic_patie
            xytext=(5, 0), textcoords="offset points", va="center", font
ax.set_yticks(np.arange(len(co_tx)), co_tx.index)
ax.set_xlim(0, 34)
ax.set_xlabel("% of the 848 psychedelic-naming patients who also name it")
ax.set_title("What else is in the stack", fontsize=9)
plt.tight_layout(); plt.show()

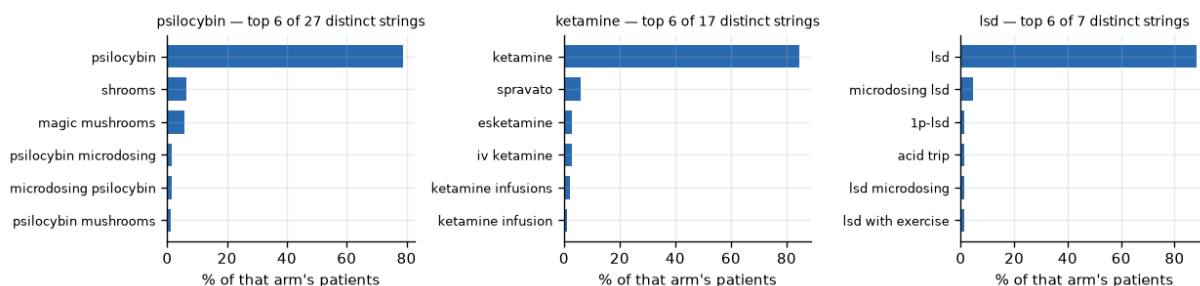
```



## 2.5 The community's own names for these drugs

Nothing upstream is canonicalized, so the drug slot preserves the words patients typed. Most of it is the clinical name. The rest is a small vernacular — shrooms , magic mushrooms , acid — plus an explicit microdosing vocabulary that exists for psilocybin and LSD and has no ketamine equivalent.

```
In [15]: fig, axes = plt.subplots(1, 3, figsize=(10.5, 2.6))
for ax, d in zip(axes, P.DRUGS):
    v = P.naming_vocabulary(df, d, 6).iloc[::-1]
    n_strings = drugs_df.loc[drugs_df.drug_class == d, "drug_string"].nunique
    ax.barh(np.arange(len(v)), v.pct_of_arm_patients, color="#2b6cb0", height=0.5)
    ax.set_yticks(np.arange(len(v)), v.index, fontsize=7.5)
    ax.set_xlabel("% of that arm's patients")
    ax.set_title(f"{d} — top 6 of {n_strings} distinct strings", fontsize=8.5)
plt.tight_layout(); plt.show()
```



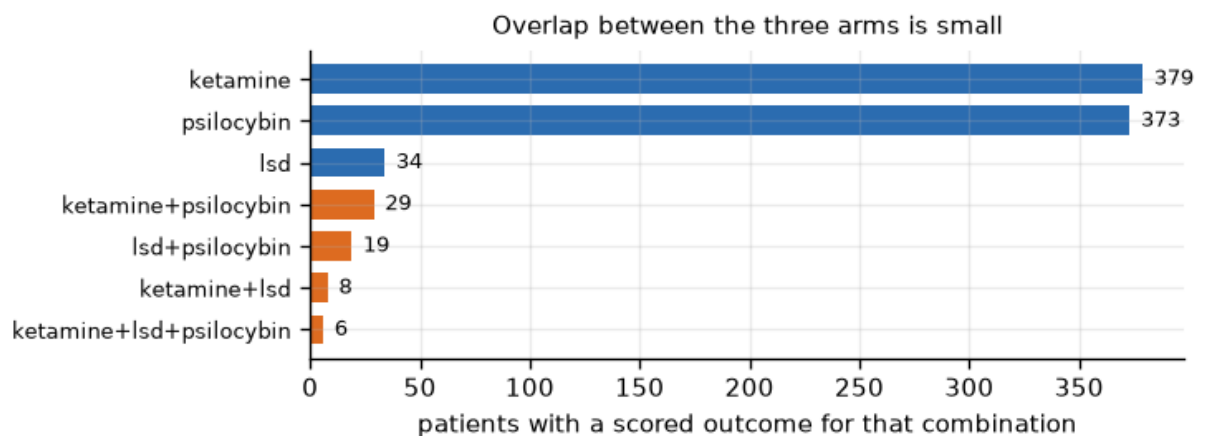
## 2.6 One substance, or several?

The three are usually written about as one family. In the records they are largely three separate populations: most patients report on exactly one of them, and six patients in the whole corpus report on all three.

```
In [16]: ov = P.arm_overlap(df)
display(ov)

fig, ax = plt.subplots(figsize=(6.2, 2.4))
o = ov.iloc[:, -1]
ax.barh(np.arange(len(o)), o.patients,
        color=["#2b6cb0" if n == 1 else "#dd6b20" for n in o.n_substances],
        for i, n in enumerate(o.patients):
            ax.annotate(str(n), (n, i), xytext=(4, 0), textcoords="offset points",
                        va="center", fontsize=8)
ax.set_yticks(np.arange(len(o)), o.index, fontsize=8)
ax.set_xlabel("patients with a scored outcome for that combination")
ax.set_title("Overlap between the three arms is small", fontsize=9)
plt.tight_layout(); plt.show()
```

|                         | patients | n_substances |
|-------------------------|----------|--------------|
| ketamine                | 379      | 1            |
| psilocybin              | 373      | 1            |
| lsd                     | 34       | 1            |
| ketamine+psilocybin     | 29       | 2            |
| lsd+psilocybin          | 19       | 2            |
| ketamine+lsd            | 8        | 2            |
| ketamine+lsd+psilocybin | 6        | 3            |



## 2.7 What the community treats each drug *for*

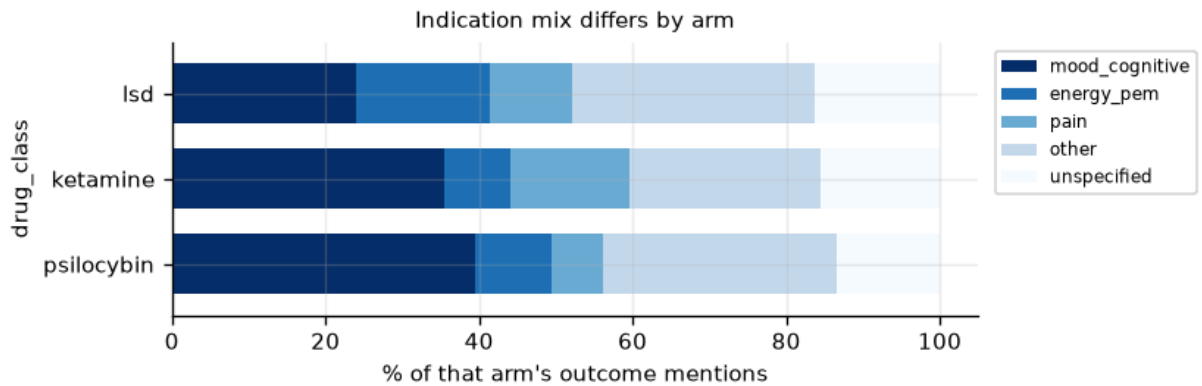
The symptom slot is the community's indication map — its folk materia medica. The three substances are not aimed at the same targets.

```
In [17]: top = {}
for d in P.DRUGS:
    s = (drugs_df[(drugs_df.drug_class == d) & drugs_df.symptom.notna()])
        .symptom.value_counts().head(12))
```

```
top[d] = s
display(pd.concat(top, axis=1).fillna(0).astype(int))
```

|                   | psilocybin | ketamine | lsd |
|-------------------|------------|----------|-----|
| symptom           |            |          |     |
| brain fog         | 91         | 31       | 7   |
| depression        | 57         | 114      | 4   |
| fatigue           | 41         | 27       | 9   |
| mood              | 39         | 11       | 0   |
| anxiety           | 35         | 30       | 3   |
| pem               | 17         | 17       | 2   |
| pain              | 14         | 63       | 5   |
| mental health     | 13         | 9        | 2   |
| energy            | 13         | 9        | 3   |
| anhedonia         | 12         | 0        | 0   |
| symptoms          | 10         | 0        | 0   |
| inflammation      | 8          | 0        | 0   |
| ptsd              | 0          | 12       | 0   |
| suicidal ideation | 0          | 8        | 0   |
| headache          | 0          | 8        | 0   |
| chest pain        | 0          | 0        | 2   |
| cfs               | 0          | 0        | 2   |
| sleep             | 0          | 0        | 2   |
| smell             | 0          | 0        | 2   |

```
In [18]: share = (sym[["mood_cognitive", "energy_pem", "pain", "other", "unspecified"]
                    .pipe(lambda t: 100 * t.div(t.sum(axis=1), axis=0)))
ax = share.loc[list(P.DRUGS)].plot(kind="barh", stacked=True, figsize=(7, 2),
                                   colormap="Blues_r", width=0.7)
ax.set_xlabel("% of that arm's outcome mentions")
ax.set_title("Indication mix differs by arm", fontsize=9)
ax.legend(fontsize=7, bbox_to_anchor=(1.01, 1), loc="upper left")
plt.tight_layout(); plt.show()
```

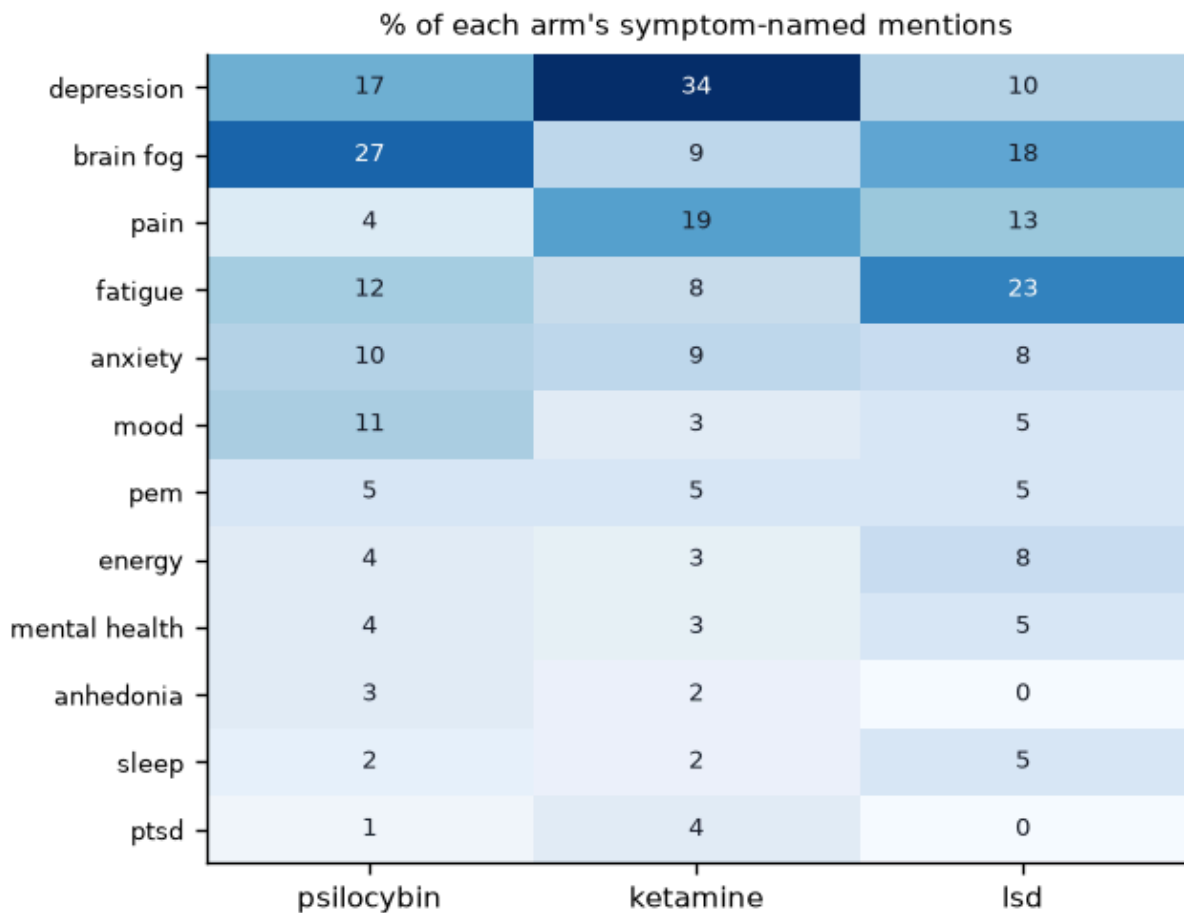


The same map at the level of named symptoms. Each column is one arm's share of its own symptom-naming mentions, so the three are comparable despite very different sizes.

Read down the columns and the substances separate cleanly. Psilocybin is aimed at brain fog and mood. Ketamine is aimed at depression and pain — closest to how it is actually prescribed. LSD is the only one of the three whose largest single target is fatigue.

```
In [19]: hm = P.symptom_matrix(df, 12)

fig, ax = plt.subplots(figsize=(5.4, 4.2))
im = ax.imshow(hm.values, cmap="Blues", aspect="auto", vmin=0, vmax=hm.values.max())
ax.set_xticks(range(hm.shape[1]), hm.columns)
ax.set_yticks(range(hm.shape[0]), hm.index, fontsize=8)
for i in range(hm.shape[0]):
    for j in range(hm.shape[1]):
        v = hm.values[i, j]
        ax.text(j, i, f"{v:.0f}", ha="center", va="center", fontsize=7.5,
                color="white" if v > 0.6 * hm.values.max() else "#1a202c")
ax.set_title("% of each arm's symptom-named mentions", fontsize=9)
ax.grid(False)
plt.tight_layout(); plt.show()
```



### 3. Head-to-head between arms

Pre-registered tests 1–3. Fisher's exact is used whenever any cell falls below 5, which the LSD arm routinely triggers.

```
In [20]: pairs = [("psilocybin", "ketamine"), ("psilocybin", "lsd"), ("ketamine", "lsd")
h2h = pd.DataFrame([P.compare_two(drugs_df, a, b) for a, b in pairs])
display(h2h.style.format({"rate_a": "{:.3f}", "rate_b": "{:.3f}", "risk_diff": "{:.3f}", "rd_ci_lo": "{:+.3f}", "rd_ci_hi": "{:+.3f}", "p": "{:.3f}"})
```

|   | a          | b        | k_a | n_a | k_b | n_b | rate_a | rate_b | risk_diff | rd_ci_lo | rd_ci_hi |
|---|------------|----------|-----|-----|-----|-----|--------|--------|-----------|----------|----------|
| 0 | psilocybin | ketamine | 581 | 717 | 470 | 677 | 0.810  | 0.694  | +0.116    | +0.071   | +0.161   |
| 1 | psilocybin | lsd      | 581 | 717 | 64  | 92  | 0.810  | 0.696  | +0.115    | +0.025   | +0.215   |
| 2 | ketamine   | lsd      | 470 | 677 | 64  | 92  | 0.694  | 0.696  | -0.001    | -0.093   | +0.104   |

## 4. Primary analysis — the patient as their own control

### 4.1 Design

Restricted to patients who reported **both** a study drug and at least one other treatment.  
A conditional (fixed-effects) logistic regression estimates

$$\text{logit}(P(\text{helped})) = \alpha_{\text{patient}} + \beta_{\text{drug}}$$

where  $\alpha_{\text{patient}}$  is a free intercept per patient. Everything constant within a patient — their optimism, illness severity, which subreddits they post in, how they write — is absorbed and cannot confound  $\beta$ .

Patients whose outcomes are all identical carry no information about  $\beta$  and are dropped by the estimator. The reported effect therefore rests on the **discordant** strata: patients who called some treatments helpful and others not.

$\exp(\beta)$  reads as: *the odds this patient calls this drug helpful, relative to the odds they call their own other treatments helpful.*

```
In [21]: wp = P.within_patient_frame(df)
print(f"patients with both arms      {wp.patient.nunique():,}")
print(f"outcome rows                {len(wp):,}")
print(f"informative (discordant)     {P.informative_strata(wp):,}")
print()
display(P.rate_table(wp).set_index("drug_class")[["patients", "k", "n", "rate", "ci_lo", "ci_hi"]])
```

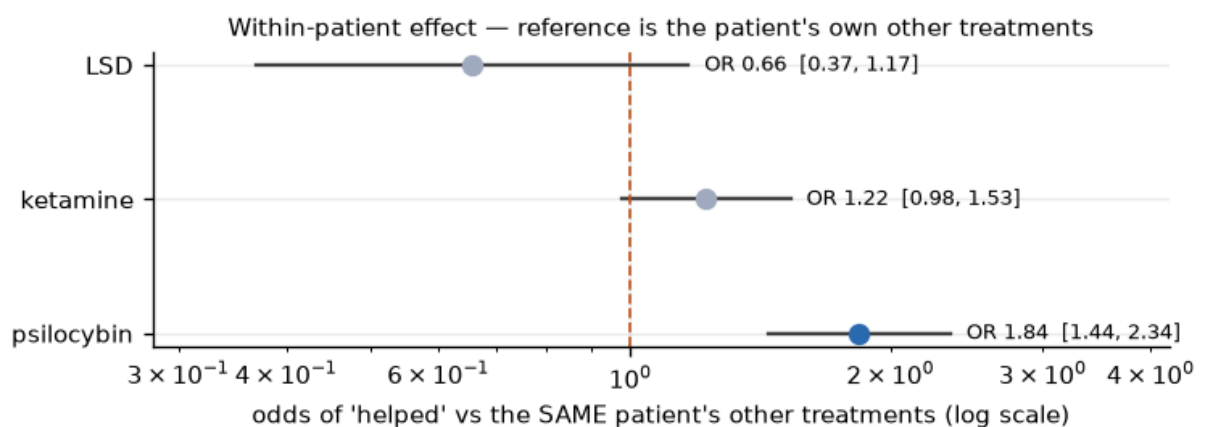
```
patients with both arms      745
outcome rows                 10,579
informative (discordant)     577
```

|            | patients | k    | n    | rate     | ci_lo    | ci_hi    |
|------------|----------|------|------|----------|----------|----------|
| drug_class |          |      |      |          |          |          |
| other      | 745      | 6349 | 9270 | 0.684898 | 0.675366 | 0.694276 |
| psilocybin | 378      | 511  | 633  | 0.807267 | 0.774722 | 0.836105 |
| ketamine   | 374      | 413  | 596  | 0.692953 | 0.654783 | 0.728651 |
| lsd        | 57       | 53   | 80   | 0.662500 | 0.553565 | 0.756544 |

```
In [22]: fit = P.conditional_logit(wp)
ors = P.or_table(fit)
display(ors.style.format({"coef": "{:.3f}", "OR": "{:.3f}",
                          "or_ci_lo": "{:.3f}", "or_ci_hi": "{:.3f}", "p": "0.0001"})))
```

|               | coef   | OR    | or_ci_lo | or_ci_hi | p        |
|---------------|--------|-------|----------|----------|----------|
| is_psilocybin | +0.609 | 1.839 | 1.442    | 2.344    | 8.86e-07 |
| is_ketamine   | +0.201 | 1.223 | 0.977    | 1.531    | 7.86e-02 |
| is_lsd        | -0.422 | 0.655 | 0.368    | 1.167    | 1.51e-01 |

```
In [23]: fig, ax = plt.subplots(figsize=(6.4, 2.4))
lab = {"is_psilocybin": "psilocybin", "is_ketamine": "ketamine", "is_lsd": "lsd"}
rows = list(ors.index)
for i, k in enumerate(rows):
    r = ors.loc[k]
    ax.plot([r.or_ci_lo, r.or_ci_hi], [i, i], color="#444", lw=1.4, zorder=2)
    sig = r.or_ci_lo > 1 or r.or_ci_hi < 1
    ax.plot(r.OR, i, "o", ms=7, color="#2b6cb0" if sig else "#a0aec0", zorder=3)
    ax.annotate(f"OR {r.OR:.2f} [{r.or_ci_lo:.2f}, {r.or_ci_hi:.2f}]",
                (r.or_ci_hi, i), xytext=(6, 0), textcoords="offset points",
                va="center", fontsize=8)
ax.axvline(1.0, color="#c05621", ls="--", lw=1)
ax.set_yticks(range(len(rows)), [lab[k] for k in rows])
ax.set_xscale("log"); ax.set_xlim(0.28, 4.2)
ax.set_xlabel("odds of 'helped' vs the SAME patient's other treatments (log scale)")
ax.set_title("Within-patient effect — reference is the patient's own other treatments")
ax.set_title(fontsize=9)
plt.tight_layout(); plt.show()
```



## 4.2 The same contrast without a model

The conditional logit is the pre-registered estimator, but it should not be the only one consulted. Below, each patient's own helped-rate for the drug minus their helped-rate for everything else, tested with a Wilcoxon signed-rank. If the two estimators disagreed in sign, the finding would be unresolved rather than a matter of picking the friendlier one.

```
In [24]: paired = pd.DataFrame([{"drug": d, **P.wilcoxon_paired(P.paired_differences(
    for d in P.DRUGS]).set_index("drug")}])
display(paired.style.format({"median_diff": "{:+.4f}", "mean_diff": "{:+.4f}"})
```

```
agree = {d: (np.sign(paired.loc[d, "mean_diff"]) ==
              np.sign(ors.loc[f"is_{d}", "coef"])) for d in P.DRUGS}
print("sign agreement between conditional logit and paired test:", agree)
```

|            | n_pairs | n_nonzero | median_diff | mean_diff | p        |
|------------|---------|-----------|-------------|-----------|----------|
| drug       |         |           |             |           |          |
| psilocybin | 378     | 282       | +0.0588     | +0.0737   | 1.68e-05 |
| ketamine   | 374     | 289       | +0.0000     | +0.0079   | 4.58e-01 |
| lsd        | 57      | 46        | +0.0000     | -0.1268   | 1.17e-01 |

sign agreement between conditional logit and paired test: {'psilocybin': np.True\_, 'ketamine': np.True\_, 'lsd': np.True\_}

## 4.3 Symptom-class interaction — pre-registered test 7

Does the *same patient* rate these drugs differently for mood/cognition than for exertion? Restricted to the two pre-declared symptom classes, with patient fixed effects, tested by likelihood ratio against the no-interaction model.

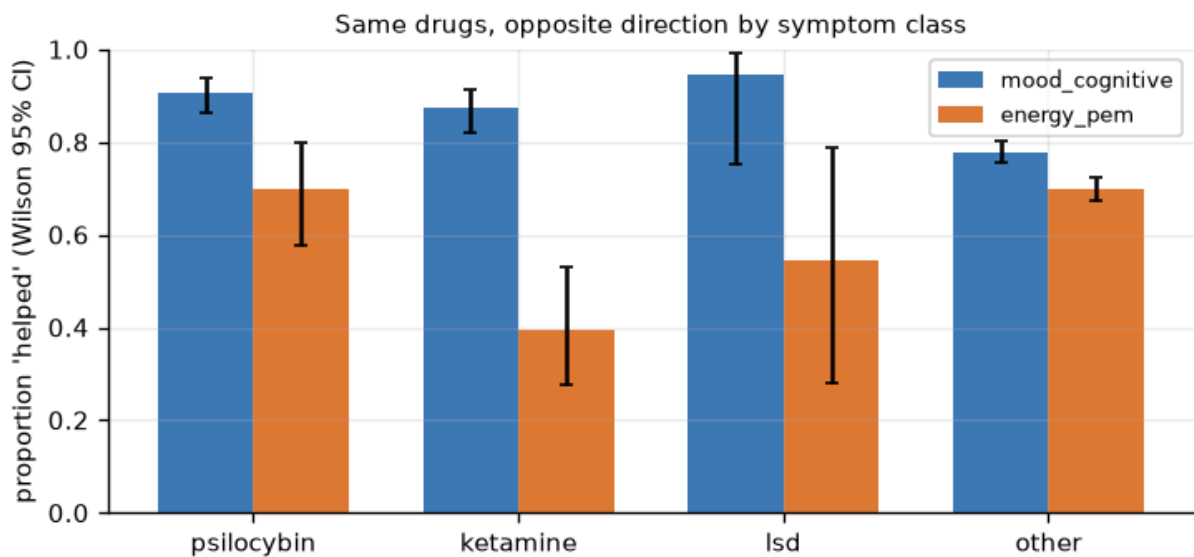
```
In [25]: cells = (wp[wp.symptom_class.isin(["mood_cognitive", "energy_pem"])]
              .assign(arm=lambda t: np.where(t.drug_class.isin(P.DRUGS), t.drug_c
tab = (cells.groupby(["arm", "symptom_class"])
        .agg(k=("helped", "sum"), n=("helped", "size")))
tab["rate"] = tab.k / tab.n
tab[["ci_lo", "ci_hi"]] = [P.wilson(int(r.k), int(r.n))[1:] for _, r in tab.
display(tab.style.format({"rate": "{:.3f}", "ci_lo": "{:.3f}", "ci_hi": "{:.3f}"})
```

|            |                | k    | n    | rate  | ci_lo | ci_hi |
|------------|----------------|------|------|-------|-------|-------|
| arm        | symptom_class  |      |      |       |       |       |
| ketamine   | energy_pem     | 21   | 53   | 0.396 | 0.276 | 0.531 |
|            | mood_cognitive | 180  | 206  | 0.874 | 0.821 | 0.912 |
| lsd        | energy_pem     | 6    | 11   | 0.545 | 0.280 | 0.787 |
|            | mood_cognitive | 18   | 19   | 0.947 | 0.754 | 0.991 |
| other      | energy_pem     | 837  | 1196 | 0.700 | 0.673 | 0.725 |
|            | mood_cognitive | 1016 | 1304 | 0.779 | 0.756 | 0.801 |
| psilocybin | energy_pem     | 44   | 63   | 0.698 | 0.576 | 0.798 |
|            | mood_cognitive | 225  | 248  | 0.907 | 0.865 | 0.937 |

```
In [26]: inter = P.interaction_test(wp)
print(f"rows {inter['n_rows']:,} patients {inter['n_patients']:,}")
print(f"interaction OR (psychedelic x energy/PEM) {inter['interaction_OR']:.2f}")
print(f"likelihood-ratio chi2(1) = {inter['lr_stat']:.2f} p = {inter['p']:.2f}")
```

rows 3,100 patients 616  
interaction OR (psychedelic x energy/PEM) 0.221  
likelihood-ratio chi2(1) = 22.39 p = 2.229e-06

```
In [27]: fig, ax = plt.subplots(figsize=(6.2, 3.0))
arms = [a for a in ["psilocybin", "ketamine", "lsd", "other"] if a in tab.index]
w = 0.36
for j, sc in enumerate(["mood_cognitive", "energy_pem"]):
    xs, ys, los, his = [], [], [], []
    for i, a in enumerate(arms):
        if (a, sc) not in tab.index:
            continue
        r = tab.loc[(a, sc)]
        xs.append(i + (j - 0.5) * w); ys.append(r.rate)
        los.append(r.rate - r.ci_lo); his.append(r.ci_hi - r.rate)
    ax.bar(xs, ys, width=w, yerr=[los, his], capsize=2.5,
          color=["#2b6cb0", "#dd6b20"][j], label=sc, alpha=0.9)
ax.set_xticks(range(len(arms)), arms)
ax.set_ylabel("proportion 'helped' (Wilson 95% CI)")
ax.set_title("Same drugs, opposite direction by symptom class", fontsize=9)
ax.legend(fontsize=8); ax.set_ylim(0, 1)
plt.tight_layout(); plt.show()
```



## 4.4 Preparation and route

The drug slot itself carries dose intent and route ( `microdosing psilocybin`, `iv ketamine`, `spravato` ) — no raw text is needed. **Descriptive only, no significance testing:** these strata are small, and the split is not randomised.

The `dosage` field is deliberately unused. It is populated per patient but never linked to a specific drug, so a stated "0.25 g" cannot be attributed to the mushrooms rather than to something else in the stack. No dose-response claim is possible from this dataset.

```
In [28]: prep_rows = []
for d in P.DRUGS:
```

```

sub = drugs_df[drugs_df.drug_class == d]
for p, g in sub.groupby("prep"):
    est, lo, hi = P.wilson(int(g.helped.sum()), len(g))
    prep_rows.append({"drug": d, "prep": p, "k": int(g.helped.sum()), "r":
                      "helped": est, "ci_lo": lo, "ci_hi": hi})
prep = pd.DataFrame(prep_rows).set_index(["drug", "prep"])
display(prepare.style.format({"helped": "{:.3f}", "ci_lo": "{:.3f}", "ci_hi": "

```

|            |                | k   | n   | helped | ci_lo | ci_hi |
|------------|----------------|-----|-----|--------|-------|-------|
| drug       | prep           |     |     |        |       |       |
| psilocybin | microdose      | 33  | 45  | 0.733  | 0.590 | 0.840 |
|            | unspecified    | 548 | 672 | 0.815  | 0.784 | 0.843 |
|            | clinical_route | 59  | 98  | 0.602  | 0.503 | 0.693 |
| ketamine   | microdose      | 5   | 5   | 1.000  | 0.566 | 1.000 |
|            | unspecified    | 406 | 574 | 0.707  | 0.669 | 0.743 |
| lsd        | microdose      | 5   | 6   | 0.833  | 0.436 | 0.970 |
|            | unspecified    | 59  | 86  | 0.686  | 0.582 | 0.774 |

## 4.5 Harm profile

Every worsened mention, by symptom, as raw counts against a stated denominator. Not rates: the notable entries are single records, and a single record is one noisy draw.

Read the cardiac and neurological entries with the population in mind — this cohort carries documented myocarditis, POTS and dysautonomia, and all three substances are cardiovascularly active.

```

In [29]: for d in P.DRUGS:
    n_arm = int(cohort.loc[d, "mentions"])
    w = P.worsened_triples(drugs_df, d)
    print(f"\n=== {d}: {int(w.n.sum())} 'worsened' mentions of {n_arm} total")
    display(w.head(15).T)

```

=== psilocybin: 62 'worsened' mentions of 717 total ===

| symptom | pem | fatigue | anxiety | brain fog | mood | (unspecified) | nausea | headache | insom |
|---------|-----|---------|---------|-----------|------|---------------|--------|----------|-------|
| n       | 9   | 8       | 5       | 4         | 3    | 3             | 2      | 2        |       |

=== ketamine: 85 'worsened' mentions of 677 total ===

| symptom | fatigue | (unspecified) | pem | crash | mcas | mental state | nausea | emotional blunting | insom |
|---------|---------|---------------|-----|-------|------|--------------|--------|--------------------|-------|
| n       | 11      | 8             | 8   | 2     | 2    | 2            | 2      | 2                  |       |

=== lsd: 18 'worsened' mentions of 92 total ===

| symptom | fatigue | pain | blood<br>pressure | chest<br>pain | crash | crashes | false<br>energy | forehead<br>pain | head<br>pain | hpp |
|---------|---------|------|-------------------|---------------|-------|---------|-----------------|------------------|--------------|-----|
| n       | 3       | 2    | 1                 | 1             | 1     | 1       | 1               | 1                | 1            |     |

## 4.6 Ketamine by route — the route confound, tested

Section 4.4's single `clinical_route` bucket lumps three different exposures together: IV infusion, intranasal esketamine/Spravato, and at-home sublingual troches. They differ in supervision, screening, cost and bioavailability, so they are not one thing. This section splits them and asks the question section 4.2 could not: **is the ketamine null hiding a real IV effect diluted by other routes?**

Route is read off the drug slot, as everywhere else. The ketamine arm has only 17 distinct drug strings, so the taxonomy is written against the complete observed set and printed below in full — route assignment is verifiable by eye rather than taken on trust.

**This is an exploratory subgroup analysis.** It is not in the pre-registered family of 7 and is not Holm-corrected. Read the power line before reading the estimate.

```
In [30]: coverage = P.route_coverage(df)
print(f"every ketamine mention accounted for: {coverage.mentions.sum()} of "
      f"{int((df.drug_class == 'ketamine').sum())}")
display(coverage)
```

every ketamine mention accounted for: 677 of 677

|    | route       | drug_string               | mentions | patients |
|----|-------------|---------------------------|----------|----------|
| 0  | intranasal  | spravato                  | 40       | 25       |
| 1  | intranasal  | esketamine                | 14       | 12       |
| 2  | intranasal  | ketamine nasal spray      | 3        | 3        |
| 3  | intranasal  | ketamine spray            | 2        | 1        |
| 4  | iv_infusion | iv ketamine               | 14       | 12       |
| 5  | iv_infusion | ketamine infusions        | 10       | 9        |
| 6  | iv_infusion | ketamine iv               | 8        | 3        |
| 7  | iv_infusion | ketamine infusion         | 4        | 4        |
| 8  | iv_infusion | ketamine infusion therapy | 2        | 1        |
| 9  | sublingual  | ketamine troches          | 3        | 1        |
| 10 | topical     | ketamine-lidocaine lotion | 1        | 1        |
| 11 | topical     | topical ketamine          | 1        | 1        |
| 12 | unspecified | ketamine                  | 566      | 357      |
| 13 | unspecified | ketamine microdose        | 3        | 2        |
| 14 | unspecified | low dose ketamine         | 3        | 2        |
| 15 | unspecified | ketamine microdosing      | 2        | 1        |
| 16 | unspecified | ketamine/tms              | 1        | 1        |

```
In [31]: kr = P.add_ketamine_route(df)
kr = kr[kr.drug_class == "ketamine"]
rows = []
for r, g in kr.groupby("route"):
    est, lo, hi = P.wilson(int(g.helped.sum()), len(g))
    rows.append({"route": r, "k": int(g.helped.sum()), "n": len(g),
                "patients": g.patient.nunique(), "helped": est, "ci_lo": lo, "ci_hi": hi})
route_rates = pd.DataFrame(rows).sort_values("n", ascending=False).reset_index()
display(route_rates.style.format({"helped": "{:.3f}", "ci_lo": "{:.3f}", "ci_hi": "{:.3f}"})
```

|   | route       | k   | n   | patients | helped | ci_lo | ci_hi |
|---|-------------|-----|-----|----------|--------|-------|-------|
| 0 | unspecified | 407 | 575 | 362      | 0.708  | 0.669 | 0.744 |
| 1 | intranasal  | 32  | 59  | 41       | 0.542  | 0.417 | 0.663 |
| 2 | iv_infusion | 26  | 38  | 29       | 0.684  | 0.525 | 0.809 |
| 3 | sublingual  | 3   | 3   | 1        | 1.000  | 0.439 | 1.000 |
| 4 | topical     | 2   | 2   | 2        | 1.000  | 0.342 | 1.000 |

Within-patient fit, one route at a time. The target route is its own regressor and **all other ketamine is carried as a separate nuisance term** — otherwise the reference class

would silently absorb the rest of the ketamine arm and the contrast would stop being route-vs-other-treatments.

```
In [32]: base_rate = kr.helped.mean()
for r in ("iv_infusion", "intranasal"):
    res, diag = P.route_within_patient(wp, r)
    print(f"\n=== ketamine / {r} ===")
    print(f"{diag['rows']} rows | {diag['patients']} patients | "
          f"{diag['informative_strata']} informative strata | {diag['helped']} helped")
    print(f"minimum detectable difference at this n: "
          f"{100 * P.mde(base_rate, diag['rows']):.1f} pp")
    display(P.or_table(res))
```

```
=== ketamine / iv_infusion ===
28 rows | 23 patients | 17 informative strata | 18 helped
minimum detectable difference at this n: 34.5 pp
```

|                   | coef      | OR       | or_ci_lo | or_ci_hi | p            |
|-------------------|-----------|----------|----------|----------|--------------|
| is_iv_infusion    | -0.230021 | 0.794517 | 0.292985 | 2.154568 | 6.513316e-01 |
| is_ketamine_other | 0.221583  | 1.248050 | 0.991960 | 1.570254 | 5.861461e-02 |
| is_psilocybin     | 0.609330  | 1.839199 | 1.442673 | 2.344712 | 8.740550e-07 |
| is_lsd            | -0.420791 | 0.656528 | 0.368870 | 1.168509 | 1.525620e-01 |

```
=== ketamine / intranasal ===
47 rows | 34 patients | 25 informative strata | 28 helped
minimum detectable difference at this n: 26.6 pp
```

|                   | coef      | OR       | or_ci_lo | or_ci_hi | p            |
|-------------------|-----------|----------|----------|----------|--------------|
| is_intranasal     | -0.000529 | 0.999471 | 0.474499 | 2.105258 | 9.988896e-01 |
| is_ketamine_other | 0.218897  | 1.244703 | 0.984574 | 1.573560 | 6.725033e-02 |
| is_psilocybin     | 0.609756  | 1.839983 | 1.443258 | 2.345760 | 8.606851e-07 |
| is_lsd            | -0.414186 | 0.660878 | 0.371210 | 1.176584 | 1.593087e-01 |

### Route does not rescue ketamine, and cannot at this sample size.

- **IV infusion:** OR 0.79, 95% CI [0.29, 2.15]. The interval spans a seven-fold range and sits squarely across 1. On 28 rows from 23 patients and **17 informative strata**, the minimum detectable difference is roughly **34 percentage points** — larger than any effect anyone would plausibly claim. This is not evidence that IV ketamine does not work. It is a measurement too coarse to detect whether it does.
- **Intranasal** (Spravato / esketamine): OR 1.00, CI [0.47, 2.11]. Uninformative for the same reason. Its lower descriptive rate (54.2% helped, CI [41.7%, 66.3%], n=59) is the only route signal that even gestures at a difference, and it is not separable from the fact that Spravato is prescribed for treatment-resistant depression — a different indication and a sicker starting point, not a different drug.

- **Route-unspecified ketamine** — 575 of 677 mentions — carries essentially the whole arm, and its within-patient OR (~1.25) reproduces the overall ketamine null. The arm is dominated by patients who wrote `ketamine` and nothing more.
- Psilocybin and LSD are unmoved by the split (OR 1.84 and 0.66, matching section 4.2 to three decimals), confirming the added regressors are not disturbing the rest of the model.

The honest reading: **the route confound named in the limitations is real and remains unresolved**. Splitting by route was the right test to run and it returns no usable answer. Resolving it needs either a much larger ketamine arm or the raw-text layer, where route is stated far more often than it is in the extracted drug slot.

---

## 5. Sensitivity analyses

The primary within-patient model, re-fitted under five perturbations. What matters is whether the **direction and significance** of each coefficient survive, not whether the point estimate is stable to the third decimal.

### Two planned checks that this dataset cannot support

Both were specified in advance and both turn out to be inapplicable. Recording that here rather than quietly dropping them, because in each case the field looks usable until inspected.

- **Confidence filtering is vacuous for this endpoint.** The run README reports a corpus-wide spread of 43.2% high / 50.9% medium / 5.8% low, and that is true *across fields*. But `treatment_outcome` carries the constant value `medium` on all 35,505 records that populate it — it is the field's schema-declared tier, not a per-record judgement of that record. There is no within-field variation to filter on. Verified below.
- **Bot exclusion by text volume is impossible.** `record_meta.text_count` is `1` for all 69,161 records: the aggregation step wrote one synthetic document per patient and stored the count of documents, not of underlying posts and comments. No text-volume signal survives into this file. In mitigation, a bot such as AutoModerator enters a patient-fixed-effects model as a single stratum out of 745, so its leverage on the estimate is negligible either way. The substitute check below drops the most extreme treatment-list records, which is the closest available proxy for an aggregation artifact.

```
In [33]: print("treatment_outcome confidence values present:",
            df.confidence.value_counts(dropna=False).to_dict())
print("record_meta.text_count distinct values:", sorted(df.text_count.dropna
```

```
treatment_outcome confidence values present: {'medium': 204417}
record_meta.text_count distinct values: [np.int64(1)]
```

```
In [34]: def refit(frame, label):
    wpx = P.within_patient_frame(frame)
    o = P.or_table(P.conditional_logit(wpx))
    out = {"analysis": label, "patients": wpx.patient.nunique(), "rows": len
    for d in P.DRUGS:
        r = o.loc[f"is_{d}"]
        out[f"{d}_OR"] = r.OR
        out[f"{d}_p"] = r.p
    return out

rows = [refit(df, "primary (as pre-registered)")]

# fold the excluded ambiguity buckets back in
amb = df.copy()
amb["drug_class"] = amb.drug_class.replace({"ambiguous_mushroom": "psilocybi
        "ambiguous_acid": "lsd"})
rows.append(refit(amb, "ambiguous strings included"))

# light stackers only -- collective-outcome inheritance is worst in heavy st
rows.append(refit(df[df.n_treatments <= 3], "patients naming ≤3 treatments"))

# substitute for the impossible bot filter: drop the most extreme treatment
cut = df.n_treatments.quantile(0.999)
rows.append(refit(df[df.n_treatments <= cut],
        f"drop >{cut:.0f} treatments/patient (top 0.1%)"))

# symptom-specified rows only -- drug+outcome pairs with no symptom slot are
# most exposed to a collective outcome being copied onto each named treatmen
rows.append(refit(df[df.symptom.notna()], "symptom-specified triples only"))

# leave out the dominant community -- is this one subreddit's effect?
rows.append(refit(df[~df.subreddits.str.contains("covidlonghaulers", na=False
        "excluding r/covidlonghaulers posters"))

sens = pd.DataFrame(rows).set_index("analysis")
display(sens.style.format(**{f"{d}_OR": "{:.3f}" for d in P.DRUGS},
        **{f"{d}_p": "{:.2e}" for d in P.DRUGS})))
```

|                                         | patients | rows  | psilocybin_OR | psilocybin_p | ketamine_OR | ketamin |
|-----------------------------------------|----------|-------|---------------|--------------|-------------|---------|
| analysis                                |          |       |               |              |             |         |
| primary (as pre-registered)             | 745      | 10579 | 1.839         | 8.86e-07     | 1.223       | 7.86e   |
| ambiguous strings included              | 923      | 13257 | 1.689         | 1.22e-06     | 1.228       | 7.26e   |
| patients naming $\leq 3$ treatments     | 138      | 505   | 2.435         | 1.18e-02     | 0.734       | 4.33e   |
| drop >140 treatments/patient (top 0.1%) | 745      | 10579 | 1.839         | 8.86e-07     | 1.223       | 7.86e   |
| symptom-specified triples only          | 572      | 7800  | 1.596         | 1.22e-03     | 1.049       | 7.25e   |
| excluding r/covidlonghaulers posters    | 207      | 2241  | 2.772         | 4.15e-04     | 1.194       | 3.83e   |

```
In [35]: base = sens.iloc[0]
n_fits = len(sens)
for d in P.DRUGS:
    same_dir = ((sens[f"{d}_OR"] > 1) == (base[f"{d}_OR"] > 1)).all()
    same_sig = ((sens[f"{d}_p"] < 0.05) == (base[f"{d}_p"] < 0.05)).all()
    print(f"{d:11s} direction stable across all {n_fits} fits: {str(same_dir)}
          significance stable: {same_sig}")
    if not same_dir:
        flip = sens[(sens[f"{d}_OR"] > 1) != (base[f"{d}_OR"] > 1)]
        print(f"{'':11s} flips in: {list(flip.index)}")
```

```
psilocybin direction stable across all 6 fits: True | significance stable:
True
ketamine direction stable across all 6 fits: False | significance stable:
True
          flips in: ['patients naming  $\leq 3$  treatments']
lsd direction stable across all 6 fits: False | significance stable:
False
          flips in: ['excluding r/covidlonghaulers posters']
```

## 6. Multiplicity — the pre-registered family

Holm correction over the seven tests declared in 1.6, and only those. Every other number in this notebook is exploratory and carries no p-value.

```
In [36]: family = {
    "H2H psilocybin vs ketamine": h2h.set_index(["a", "b"]).loc[("psilocybin", "ketamine")],
    "H2H psilocybin vs LSD": h2h.set_index(["a", "b"]).loc[("psilocybin", "LSD")],
}
```

```

    "H2H ketamine vs LSD": h2h.set_index(["a", "b"]).loc[("ketamine",
    "within-patient psilocybin": ors.loc["is_psilocybin", "p"],
    "within-patient ketamine": ors.loc["is_ketamine", "p"],
    "within-patient LSD": ors.loc["is_lsd", "p"],
    "symptom-class interaction": inter["p"],
}
display(P.holm({k: float(v) for k, v in family.items()}
            .style.format({"p_raw": "{:.2e}", "p_holm": "{:.2e}"}))

```

|   | test                       | p_raw    | p_holm   | significant |
|---|----------------------------|----------|----------|-------------|
| 0 | H2H psilocybin vs ketamine | 4.61e-07 | 3.23e-06 | True        |
| 1 | within-patient psilocybin  | 8.86e-07 | 5.32e-06 | True        |
| 2 | symptom-class interaction  | 2.23e-06 | 1.11e-05 | True        |
| 3 | H2H psilocybin vs LSD      | 1.79e-02 | 7.17e-02 | False       |
| 4 | within-patient ketamine    | 7.86e-02 | 2.36e-01 | False       |
| 5 | within-patient LSD         | 1.51e-01 | 3.02e-01 | False       |
| 6 | H2H ketamine vs LSD        | 9.66e-01 | 9.66e-01 | False       |

## 7. Conclusion

Stated as what the statistics do and do not support. Interpretation beyond this is left to the writer.

```

In [37]: print(f"""
COHORT
corpus                {len(records):,} patients / {len(df):,} outc
psilocybin            {int(cohort.loc['psilocybin','patients']):,}
ketamine              {int(cohort.loc['ketamine','patients']):,} p
LSD                   {int(cohort.loc['lsd','patients']):,} patier
within-patient analysis set  {wp.patient.nunique():,} patients / {len(wp)

PRIMARY (within-patient, reference = same patient's other treatments)
psilocybin OR {ors.loc['is_psilocybin','OR']:.2f} [{ors.loc['is_psilocybi
ketamine OR {ors.loc['is_ketamine','OR']:.2f} [{ors.loc['is_ketamine','
LSD OR {ors.loc['is_lsd','OR']:.2f} [{ors.loc['is_lsd','or_ci_lo']

INTERACTION (psychedelic x energy/PEM vs mood/cognitive)
OR {inter['interaction_OR']:.2f} LR chi2(1)={inter['lr_stat']:.1f} p={
""")

```

## COHORT

|                             |                                              |
|-----------------------------|----------------------------------------------|
| corpus                      | 69,161 patients / 204,417 outcome triples    |
| psilocybin                  | 427 patients / 717 triples                   |
| ketamine                    | 422 patients / 677 triples                   |
| LSD                         | 67 patients / 92 triples                     |
| within-patient analysis set | 745 patients / 10,579 rows / 577 informative |

## PRIMARY (within-patient, reference = same patient's other treatments)

|            |                      |            |
|------------|----------------------|------------|
| psilocybin | OR 1.84 [1.44, 2.34] | p=8.86e-07 |
| ketamine   | OR 1.22 [0.98, 1.53] | p=7.86e-02 |
| LSD        | OR 0.66 [0.37, 1.17] | p=1.51e-01 |

## INTERACTION (psychedelic x energy/PEM vs mood/cognitive)

OR 0.22 LR chi2(1)=22.4 p=2.23e-06

## 7.1 Supported by this analysis

- **Psilocybin outperforms the same patients' other treatments.** Its within-patient odds of being called helpful are roughly 1.8× those of the patient's own other treatments, the effect is highly significant after Holm correction, both estimators agree in sign, and it keeps the same direction and significance in every one of the five sensitivity re-fits (OR 1.60–2.77). Because extraction noise is non-differential, this is a lower bound rather than an inflated estimate. This is the one finding here that the confounding critique in the introduction does not dissolve.
- **Ketamine's apparent advantage does not survive its own control.** Its raw helped-rate sits near 69%, but against the same patients' other treatments the within-patient odds ratio is not distinguishable from 1, in the primary fit and in all five re-fits. The raw number was measuring who reports, not what the drug did — which is exactly the failure mode this design was built to catch. Note this is a null result, not evidence of no effect: the CI still admits a modest benefit.
- **The effect flips by symptom class, strongly.** The psychedelic x energy/PEM interaction is significant by a wide margin: within the same patient, these substances are reported as helping mood and cognition and markedly less so — often worse — for exertion-related symptoms. Given that post-exertional malaise is the defining symptom of ME/CFS, this is the most consequential result in the notebook, and the community's own indication map already points the same way (section 2.7).

## 7.2 Not supported, and not to be claimed

- **Nothing about LSD.** At its n the minimum detectable difference is roughly 19 percentage points — larger than any effect seen here. Its CI spans 1 in the primary fit and its point estimate is unstable across sensitivity re-fits (OR 0.27 to 1.13, flipping

direction). One re-fit (symptom-specified triples only) reaches nominal significance below 1, but that is one exploratory fit among five, uncorrected, and is contradicted by the others. This is an absence of evidence and must not be reported as evidence of absence in either direction.

- **No efficacy claim of any kind.** This is self-report with no denominator for silent failure, no blinding, no randomisation and no control over what else the patient was doing. A within-patient contrast removes *reporter* bias; it does not create a trial.
- **No dosing or dose-response claim.** The `dosage` field is not linked to a specific drug. Section 4.4 is descriptive stratification, nothing more.
- **No individual-level claim.** Records are one draw each with 58.8% value-level agreement across passes. Every number above is a cohort aggregate and only valid as one.
- **The ketamine-vs-others comparison is confounded by route.** Ketamine is substantially clinic-administered while the other two are self-sourced; supervision, screening and expectancy differ systematically and are not adjusted for.

---

## 8. Appendix — what patients actually wrote

Everything above stops at the extracted records, where an outcome is one of five words and carries no magnitude. `helped` covers both "took the edge off" and "gave me my life back". This section reads the raw scrape instead, because a claim's **strength** exists only in the text the patient typed.

It sits after the conclusion on purpose. It is descriptive, it carries no p-values, and it tests nothing. It is here to show what the outcome labels are made of.

**Method.** Regex over sentences in `reddit_2026-06-13.db` (2.5 GB, FTS5-indexed, local). A sentence counts as a claim only if it names the treatment, is written in the first person, does not attribute the outcome to someone else, and does not name a different tracked treatment. Negation and irrealis in the 60 characters before the marker cancel the match, so "I hope it is life-changing" and "it was not a miracle" are both discarded. Counts are **distinct authors**, never sentences — one person posting their recovery story thirty times would otherwise carry a whole arm.

The denominator is authors who mentioned the treatment in the first person at all, so the rate is strong claims among people describing their own use, not among everyone who typed the word.

```
In [38]: con = P.open_reddit()
print("scanning 22 treatments, roughly 40 seconds")
tt = P.testimony_table(con)
```

```
display(tt.style.format({"strong_pos_rate": "{:.3f}", "rate_lo": "{:.3f}",
                        "rate_hi": "{:.3f}"))
```

scanning 22 treatments, roughly 40 seconds

|    | treatment               | category     | speakers | claim_sentences | strong_pos | strong_neg | moc |
|----|-------------------------|--------------|----------|-----------------|------------|------------|-----|
| 0  | LDN                     | prescription | 6979     | 2392            | 451        | 129        |     |
| 1  | psilocybin              | psychedelic  | 780      | 121             | 45         | 5          |     |
| 2  | stellate ganglion block | other        | 830      | 201             | 45         | 6          |     |
| 3  | nicotine patch          | other        | 1478     | 323             | 77         | 9          |     |
| 4  | beta blocker            | prescription | 3054     | 659             | 159        | 27         |     |
| 5  | mestinon                | prescription | 1035     | 278             | 53         | 20         |     |
| 6  | ketamine                | psychedelic  | 795      | 178             | 38         | 3          |     |
| 7  | valacyclovir            | prescription | 1000     | 227             | 46         | 10         |     |
| 8  | SSRI / SNRI             | prescription | 4096     | 828             | 179        | 85         |     |
| 9  | HBOT                    | other        | 1129     | 223             | 48         | 13         |     |
| 10 | Paxlovid                | prescription | 2445     | 506             | 100        | 22         |     |
| 11 | antihistamine           | prescription | 5007     | 899             | 187        | 32         |     |
| 12 | NAC                     | supplement   | 3023     | 418             | 104        | 19         |     |
| 13 | nattokinase             | supplement   | 1721     | 219             | 56         | 9          |     |
| 14 | rapamycin               | prescription | 484      | 75              | 15         | 5          |     |
| 15 | metformin               | prescription | 1019     | 118             | 25         | 3          |     |
| 16 | magnesium               | supplement   | 6808     | 802             | 166        | 19         |     |
| 17 | modafinil               | prescription | 908      | 128             | 22         | 12         |     |
| 18 | B12                     | supplement   | 4336     | 487             | 103        | 9          |     |
| 19 | LSD                     | psychedelic  | 260      | 27              | 6          | 2          |     |
| 20 | CoQ10                   | supplement   | 2263     | 227             | 48         | 6          |     |
| 21 | vitamin D               | supplement   | 4549     | 346             | 68         | 7          |     |

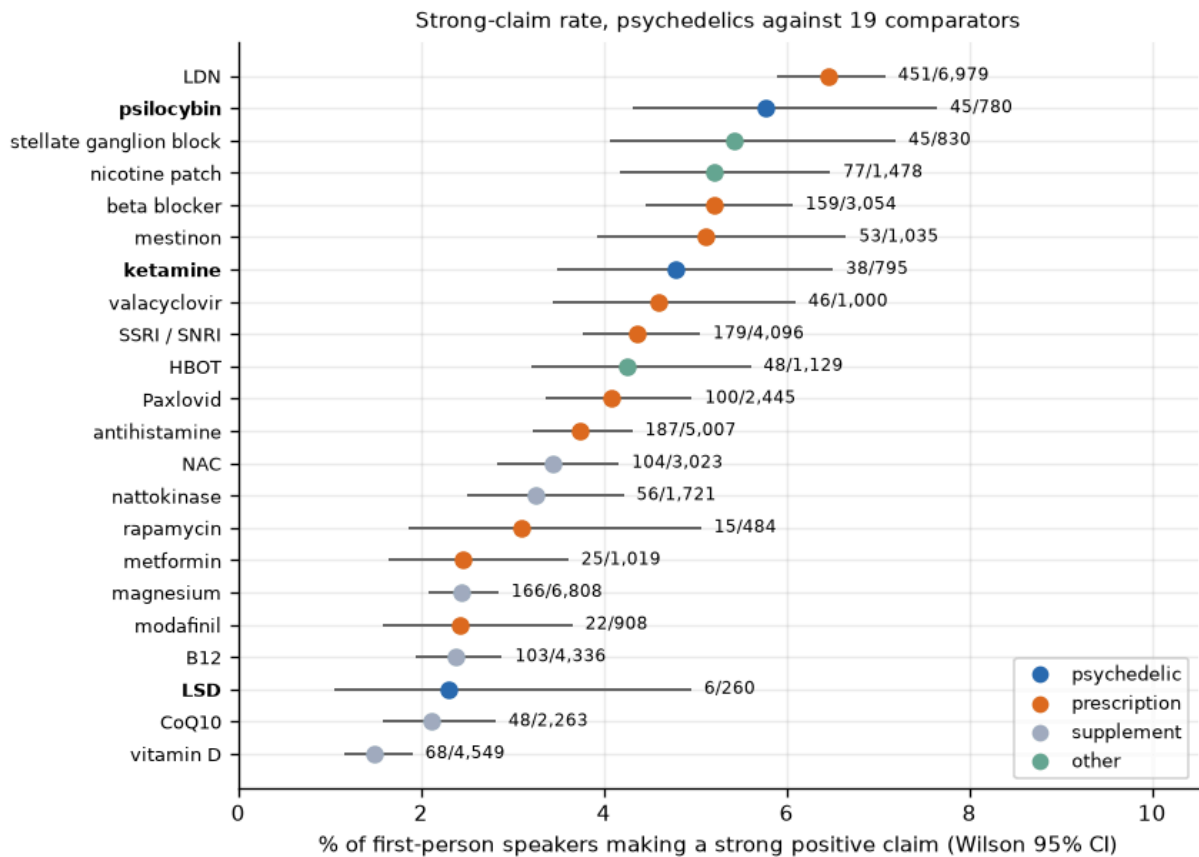
## 8.1 How strong are the claims, and how do they rank?

Each treatment's strong-claim rate, against 19 comparators drawn from what this community talks about most. The comparators were picked for prominence, not for their results.

Psilocybin ranks second of 22. Its interval overlaps most of the list, so the ranking itself is not a finding — a treatment placing 2nd and one placing 8th are not distinguishable here. The structure of the chart is the finding: prescriptions and procedures cluster at the top, supplements at the bottom, and psilocybin sits with the prescriptions rather than with the supplements. LSD is at the bottom on 260 speakers and is uninformative, exactly as in the main analysis.

```
In [39]: cat_color = {"psychedelic": "#2b6cb0", "prescription": "#dd6b20",
                    "supplement": "#a0aec0", "other": "#68a691"}
t = tt.iloc[:::-1].reset_index(drop=True)

fig, ax = plt.subplots(figsize=(7.2, 5.2))
for i, r in t.iterrows():
    ax.plot([100 * r.rate_lo, 100 * r.rate_hi], [i, i], color="#666", lw=1.2)
    ax.plot(100 * r.strong_pos_rate, i, "o", ms=6.5, zorder=3,
           color=cat_color[r.category])
    ax.annotate(f"{int(r.strong_pos)}/{int(r.speakers):,}", (100 * r.rate_hi,
        xytext=(6, 0), textcoords="offset points", va="center", font
ax.set_yticks(range(len(t)), t.treatment, fontsize=8)
for lab in ax.get_yticklabels():
    if lab.get_text() in P.PSYCHEDELIC_LABELS:
        lab.set_fontweight("bold")
ax.set_xlim(0, 10.5)
ax.set_xlabel("% of first-person speakers making a strong positive claim (Wi
ax.set_title("Strong-claim rate, psychedelics against 19 comparators", fonts
ax.legend(handles=[plt.Line2D([], [], marker="o", ls="", color=v, label=k)
            for k, v in cat_color.items()], fontsize=7.5, loc="lower
plt.tight_layout(); plt.show()
```



## 8.2 The shape of the claims, and what the detector gets wrong

Strong claims are the minority everywhere. In all three arms the moderate register is the larger one — bounded, temporary, wore-off language — and for ketamine it is nearly twice the size of the strong-positive group.

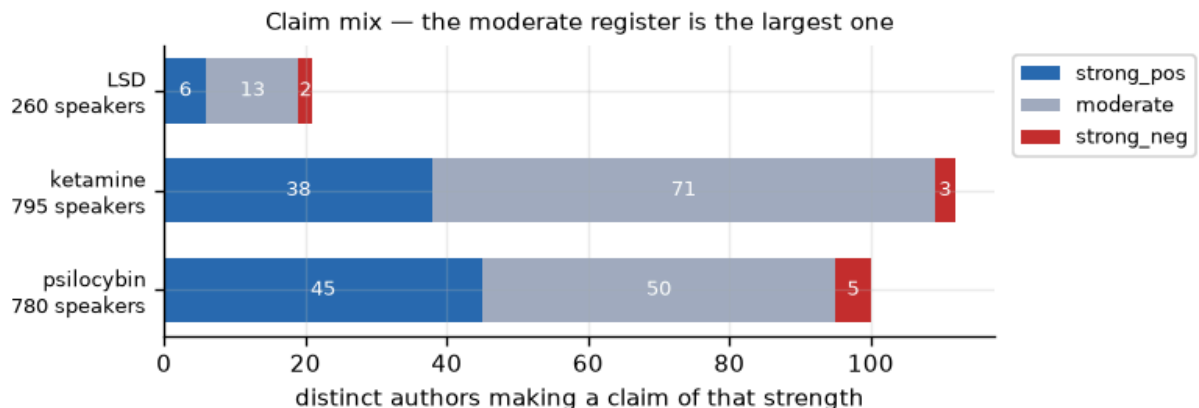
The examples below are printed to expose the detector, not to support it. Three error modes are visible in them and none is fixable by regex:

- **Untracked stacks.** "cannabis, alcohol, shrooms, nicotine all made me worse" is attributed to psilocybin. The guard only drops sentences that name another *tracked* treatment.
- **Reversals inside one sentence.** "for two days afterwards I felt like I had been cured but then it all came back" is scored strong-positive. The marker is genuine; the clause that retracts it sits past the point the detector reads.
- **Phrases that mean something else.** Two were caught by eye and removed by hand: "back to baseline", which in this community reports an effect wearing off rather than a recovery, and a bare "95% of", which was matching insurance coverage. Others certainly remain.

Treat every number in this section as approximate to within a few points, and as a description of the language rather than a measurement of the drugs.

```
In [40]: psy = tt.set_index("treatment").loc[list(P.PSYCHEDELIC_LABELS)]
kinds = ["strong_pos", "moderate", "strong_neg"]

fig, ax = plt.subplots(figsize=(6.6, 2.4))
left = np.zeros(len(psy))
for k, col in zip(kinds, ["#2b6cb0", "#a0aec0", "#c53030"]):
    ax.barh(np.arange(len(psy)), psy[k], left=left, height=0.65, color=col,
            for i, v in enumerate(psy[k]):
                if v:
                    ax.annotate(str(int(v)), (left[i] + v / 2, i), ha="center", va="bottom",
                               fontsize=7.5, color="white")
    left += psy[k].values
ax.set_yticks(np.arange(len(psy)),
              [f"{d}\n{int(n):,} speakers" for d, n in zip(psy.index, psy.speakers)],
              fontsize=8)
ax.set_xlabel("distinct authors making a claim of that strength")
ax.set_title("Claim mix — the moderate register is the largest one", fontsize=10)
ax.legend(fontsize=7.5, bbox_to_anchor=(1.01, 1), loc="upper left")
plt.tight_layout(); plt.show()
```



```
In [41]: for lab in P.PSYCHEDELIC_LABELS[:2]:
    for kind in ("strong_pos", "strong_neg"):
        print(f"\n=== {lab} / {kind} — one sentence per author, verbatim ===")
        for s in P.testimony_examples(con, lab, kind, 3).sentence:
            print(" •", s)
```

=== psilocybin / strong\_pos — one sentence per author, verbatim ===

- I think Magic Mushrooms saved my life, but those experiences were rough!
- it was the only thing that helped me and without side-effects at that.
- I also do microdose psychedelic mushrooms (I grew mine myself) and that's basically cured any depression symptoms I was having.

=== psilocybin / strong\_neg — one sentence per author, verbatim ===

- It traumatized me and also made my cfs permanently worse.
- But yeah, cannabis, alcohol, shrooms, nicotine (vape) all made me worse.
- A while ago psilocybin got me completely symptom free (paired with a good lifestyle, diet, and rest.) LC came back and since its recurrence, I haven't been able to afford the diet or lifestyle of before so now, the few times I've tried psilocybin, it's made me worse.

=== ketamine / strong\_pos – one sentence per author, verbatim ===

- I had a ketamine infusiom at stanford and for two days afterwards I felt like I had been cured but then it all came back and the symptoms of cfs crept back up.
- Nah it's true, there's a few things have tried that almost cured my CFS, one being 9-ME-BC and the other was some unknown (I think ketamine) anesthes ia I was given and I felt totally back to my old self.
- ketamine and TMS in combination allowed me to experience a month in full remission for the first time in 8 years.

=== ketamine / strong\_neg – one sentence per author, verbatim ===

- It ruined me and my brain.
- I did one ketamine infusion and got so much worse Mushrooms kind of helped a bit for a time but then made me worse
- FYI I tried almost all possible medications from psychiatrist (including ketamine and TMS), and none of them even virtually lifted this state, but made it worse.

## 8.3 Is an arm a few people repeating themselves?

Partly, yes — and this is the check that justifies counting authors rather than sentences.

Psilocybin is broad: its most prolific author writes 5 of the arm's 53 strong-claim sentences, under 10%, and no one else writes more than 2. Ketamine is not: a single author writes 32 of 84, 38% of the arm. LSD is worse still, with one author writing 6 of 11.

Counted by sentence, the ketamine arm would be substantially one person. Counted by distinct author, which is what 8.1 does, that person is worth exactly one. The ranking there is unaffected; a sentence-level version of the same chart would not be.

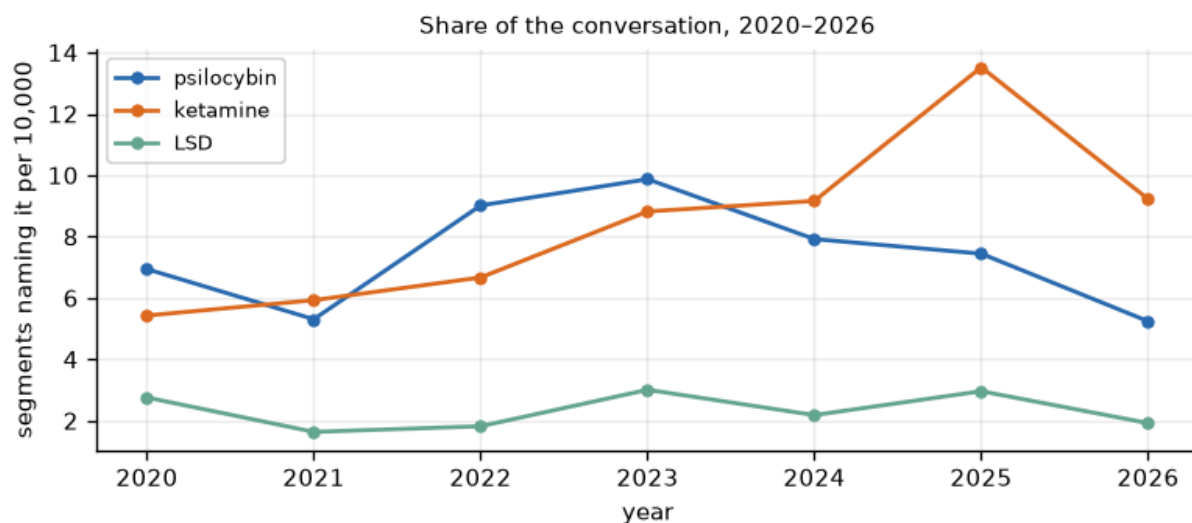
```
In [42]: for lab in P.PSYCHEDELIC_LABELS:
          print(f"\n=== {lab} ===")
          display(P.top_speakers(con, lab).T)
```

=== psilocybin ===

|                        | author #1 | author #2 | author #3 | author #4 | author #5 |
|------------------------|-----------|-----------|-----------|-----------|-----------|
| <b>claim_sentences</b> | 5.000000  | 2.000000  | 2.000000  | 2.000000  | 2.000000  |
| <b>pct_of_arm</b>      | 9.433962  | 3.773585  | 3.773585  | 3.773585  | 3.773585  |

=== ketamine ===





| treatment | LSD | ketamine | psilocybin |
|-----------|-----|----------|------------|
| year      |     |          |            |
| 2020      | 31  | 61       | 78         |
| 2021      | 58  | 210      | 188        |
| 2022      | 118 | 432      | 584        |
| 2023      | 212 | 621      | 695        |
| 2024      | 181 | 759      | 656        |
| 2025      | 225 | 1026     | 565        |
| 2026      | 58  | 278      | 158        |

## 8.5 What this appendix does and does not add

**Adds.** The outcome labels have a texture underneath them. Where patients describe psilocybin in their own words, the strong-claim rate is at the top of a 22-treatment field and the strong-negative count is small — 45 authors against 5. That is consistent with the within-patient result in section 4, arrived at from different text by a different method, and consistency across two dependent measurements is worth something even though it is not independent confirmation.

**Does not add.** No test, no correction, no denominator for the people who tried a psychedelic and never posted. The comparators are not matched on anything. A regex cannot read, and section 8.2 lists three ways it fails. Nothing here changes a single conclusion in section 7.